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Detection of Cyberbullying using Machine Learning

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ABSTRACT

Cyberbullying is a form of bullying in which technology is used as a medium to bully someone. As the new boom of the Internet and other social media platforms are increasing, the number of users is also increasing and the main users of social media are mostly teens and young adults. As much as these social media platforms are used for getting new information and for entertainment, it is more prone for bullies to use these networks as vulnerable to attacks against victims. Due to the increase in cyberbullying on victims, it is in need to develop a suitable method for the detection and prevention of cyberbullying. A growing body of work is emerging on automated approaches to cyberbullying detection. These approaches utilize machine learning and natural language processing techniques to identify the characteristics of a cyberbullying exchange and automatically detect cyberbullying by matching Textual data. The main objective of this project is to detect cyberbullying by matching both Image and Textual data. The test cases and are used to classify the dataset and detect the bullying. Machine learning techniques are used to efficiently predict and detect cyberbullying.

Keywords— Cyberbullying, Support Vector Machine, k nearest neighbor, Naïve Bayes, Decision tree, Neural Network

1. INTRODUCTION

Social networking sites are extraordinary instruments for interfacing with individuals. In any case, as Social networking sites have become across the board, individuals are finding unlawful and exploitative approaches to utilize these networks. We see that individuals, particularly teenagers what's more, youthful grown-ups, are finding better approaches to menace each other over the Internet. Nearly 25% of guardians in an investigation led by Symantec announced that, as far as anyone is concerned, their kid has been associated with a cyberbullying occurrence. Cyberbullying is a type of harassment utilizing electronic methods. Cyberbullying is known as Internet harassing too. It has gotten progressively

normal, particularly among adolescents. Cyberbullying is the point at which somebody, menace or bother others on web-based social networking locales. Unsafe tormenting conduct can incorporate posting gossipy tidbits, dangers, sexual comments, a victim's very own data, or pejorative names (i.e., detest discourse). Tormenting or on the other hand badgering can be recognized by rehashed conduct and an expectation to hurt. Casualties may have lower confidence, expanded self-destructive ideation, and an assortment of passionate reactions, counting being frightened, disappointed, furious, and discouraged.

The figure [1] shows a case of cyberbullying where an individual gets corrupting remarks on his/her post. Awareness in the United States has risen in the 2010s, due in part to prominent cases. A few US states and different nations have laws explicit to cyberbullying. In any case, what use are these laws if the cyberbullying cases are expanding. Past work has been centered on the location of cyberbullying after it previously occurred. So we have put forth an attempt to identify cyberbullying and alert the specialists about these at the beginning time. Our calculation utilizes a mix of outward appearance recognition and Natural Language Processing (NLP) to recognize cyberbullying. Facial expression detection can be utilized to distinguish and catch the sort of feeling in an image posted on and social Cyberbullying Detection media. This joined with the handling of remarks on the particular post can give us a solid end concerning whether there is harassing of any sort.



Fig. 1: Example of Cyber Bullying

Thus, many machine-learning techniques are applied to detect cyberbullying.

2. LITERATURE SURVEY

John Hani, Mohamed Nashaat, Mostafa Ahmed, Zeyad Emad, Eslam Amer et al. [1] They have given the supervised machine learning ways to detect and prevent cyberbullying. They have concentrated mainly on approaches like SVM and NN. The performance of SVM and Neural Network classifiers are compared on both TFIDF and sentiment analysis feature extraction methods. The approach they've used contains three steps, those are preprocessing, features extraction and classification. They have split the dataset in a ratio of 0.8 and 0.2 for training and testing. They have used the different n-gram language models, mainly 2-gram, 3-gram and 4-gram during the evaluation of the model. The SVM model gives an accuracy of 90.3% whereas NN performs better and gives an accuracy of 92.8%.

Kelly Reynolds, April Kontostathis and Lynne Edwards et al. [2] They have used machine learning techniques like C4.5 decision tree learning and instance-based learning. They have used these techniques to detect language patterns used by the bullies. Also, they have compared the NUM training set and NORM training set, except for the SMO algorithm. They have also used the J48 algorithm to get the overall accuracy, which is 81.7%, and it is a language-based method for detecting cyberbullying. It records the percentage of a curse and insults words within the post. The accuracy of detecting the cyberbullying is identified as 78.5%.

Cynthia Van Heel, Gilles Jacobs, Chris Emmery, Bart Desmet, Els Lefever, Ben Verhoeven, Guy De Pauw, Walter Daelemans, VeÅronique Hoste et al. [3] They have made use of the linear support vector machines for automatic detection of cyberbullying in social media. They have preferred machine learning techniques over the rule-based modeling for the detection of cyberbullying. They have spoken about supervised and semi-supervised learning and also about n-gram. They have given information about data collection and annotation. They have used binary classification experiments with the help of a Support Vector Machine (SVM) for the detection of cyberbullying. The unoptimized word n-gram-based classifier and keyword-matching system serve as baselines for comparison. After feature and hyperparameter optimization of their models, a maximum F1 score of 64.32% and 58.72% was obtained for English and Dutch (two languages trained for the detection of cyberbullying in this paper) respectively.

Ghada M. Abaido et al. [4] Mainly focused on the cyberbullying among university students in an Arab community. Calls for smartphone applications, stricter legal actions and proactive measures are discussed here. They have also mentioned about direct reporting and have made use of the Cronbach's alpha formula. With the help of both Statistical Package for the Social Sciences (SPSS) and the alpha formula they have reached the result i.e., $\alpha = 0.718$. A tabled comparison for all the criteria considered for cyberbullying is given in this paper.

Elizabeth Byrne and Lauren Pfeifer et al. [5] Here they are talking about the school nursing practice and how it can be practiced and used for the prevention of cyberbullying. The outcomes area for this integrative survey is isolated into two phases. The principal organizer is to order, outline, and combine the key discoveries from the six articles. The second

arrangement is to introduce the key examples and topics that were separated during the information examination. School medical attendants should be knowledgeable in perceiving web-based social networking, understanding the wording; also, really utilizing the normal Internet based life applications. School medical attendants are in the position to advance a positive school culture and work with understudies, other school faculty, and guardians to decrease cyberbullying.

Mohammed Ali Al-Garadi, Mohammed Rashid Hussain, Nawsher Khan, Ghulam Murtaza, Henry Friday Nweke, Ihsan Ali, Ghulam Mujtaba, Haruna Chiroma, Hasan Ali Khattak and Abdullah Gani et al. [6] This paper gives bits of knowledge on the general procedure for cyberbullying recognition and in particular reviews the approach. They have fundamentally concentrated on feature selection algorithms and then using various machine learning algorithms for prediction of cyberbullying behaviors. The most widely recognized methods for developing cyberbullying forecast models is to utilize a book classification approach that includes the development of AI classifiers from named content cases. They have also talked about Feature selection algorithms being rarely adopted in state-of-the-art research to perform cyberbullying prediction on SM websites via machine learning. They have given four aspects of cyberbullying detection and they are data collection, feature engineering, construction of cyberbullying detection model, and evaluation of constructed cyberbullying detection models.

Maral Dadvar and Kai Eckert et al. [7] have spoken about deep learning-based models for the detection of cyberbullying on social media and also by overcoming the conventional models. They have likewise moved and assessed the presence of the models prepared on one stage to another stage. Comparison between the deep learning-based models and the machine learning models and thereby concluding that deep learning-based models outperform the machine learning models. They have given a brief about Deep Learning, Online Bullying, Neural Networks, Social Networks and Transfer Learning. They have used four different DNN models for detection for cyberbullying; they are Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), Bidirectional LSTM (BLSTM) and BLSTM with attention. They have spoken about word embedding is the process of representing each word as a real value vector.

Rui Zhao and Kezhi Mao et al. [8] They have spoken about the Machine learning techniques that make automatic detection of bullying messages in social media possible. They have talked about the new method named Semantic-Enhanced Marginalized Denoising Auto-Encoder (smSDA) which is developed via the semantic extension of the popular deep learning model stacked denoising autoencoder. The semantic expansion comprises of semantic dropout commotion and sparsity limitations, where the semantic dropout clamor is structured dependent on space information and the word inserting strategy. Their proposed strategy can abuse the concealed component structure of harassing data and get familiar with a hearty and discriminative portrayal of content. They have spoken about Text Mining, Representation Learning, Stacked Denoising Autoencoders and Word Embedding.

Semiu Salawu, Yulan He, and Joanna Lumsden et al. [9] have categorized the present approaches into 4 main classes, they are supervised learning, lexicon-based, rule-based and mixed-

initiative approaches. They have talked about SVM and Naïve Bayes in a supervised learning method for the detection of cyberbullying. Rules-based methodologies coordinate content to predefined rules to distinguish harassing and mixed-initiatives approaches to consolidate human-based prevailing upon at least one of the previously mentioned approaches.

Huma Jameel, Neelam Duhan and Sana Shakeel et al. [10] this paper is an outline of cyberbullying which happens overwhelmingly on long range interpersonal communication locales and issues and difficulties in distinguishing cyberbullying. The theme introduced right now with a presentation on cyberbullying: definition, classes, and jobs. At that point, in the conversation of cyberbullying recognition, practical information sources, highlights and order systems utilized are surveyed. Natural Language Processing (NLP) and Machine Learning techniques are the noticeable methodologies used to recognize tormenting catchphrases inside the corpus. At long last, issues and difficulties in cyberbullying, identification is featured and talked about.

Amanpreet Singh and Maninder Kaur et al. [11] this paper mainly concentrates on content-based learning. Content-based learning is one of the obvious regions of information mining that manages the acknowledgment and assessment of menace substance as a rule introduced at Internet based life. These works present a precise basic examination to collect, research, capture and investigate the examples and efficiently study holes. The investigation depicts a far-reaching efficient writing survey of procedures proposed in the field of substance-based cybercrime. The overview identifies with a few information pre-handling procedures, content-based elements, Machine Learning techniques, online long range informal communication datasets and assessment parameters utilized in the setting of identifying content-based cybercrime.

L. P. D. Bosque and S. E. Garza et al. [12] The utilization of the Machine Learning techniques for consequently

distinguishing cases of cyberbullying; an essential assignment inside this cyberbullying identification comprises of aggressive text detection. They have given the different computational techniques for carrying out the task of detecting cyberbullying, either as a classification or as a regression problem, and their results suggest that a key feature is the identification of profane words.

Konstantia Barmatsalou, Tiago Cruz, Edmundo Monteiro and Paulo Simoes et al. [13] This paper presents a joined criminal profiling and suspicious example recognition technique for two crimes with moderate to substantial inclusion of cell phones which are cyberbullying and low level drug dealing. Neural and Neurofuzzy techniques are applied on a half and half unique and reenacted dataset. They have also made use of Fuzzy Logic and back propagation perceptron.

Mohd Fazil and Muhammad Abulaish et al. [14] have discussed the hybrid approach for detecting cyberbullying. These approaches are an amalgamation of feature categories such as metadata-based, content-based and interaction-based features. They have used three machine-learning techniques and these classifiers are random forest, decision tree and Bayesian network on a real dataset. Interaction-based and community-based features are also discussed here.

Tracy Vaillancourt, Robert Faris and Faye Mishna et al. [15] they have reviewed the ongoing writing on cyberbullying and its impacts on deceived youth, distinguishing key focuses. The highlights incorporate the inescapable, ceaseless nature of cyberbullying and the capacity to rapidly contact enormous crowds. The potential for obscurity and the related separation managed by screens and gadgets contrasted with face to face collaboration permits the savagery of cyberbullying to go unchecked. They have finished up by calling for schedule screening of harassing by social insurance suppliers who manage pediatric populace.

Table 1: Summary of the literature survey

S no.	Authors	Methodology	Algorithms used
1.	John Hani, Mohamed Nashaat, Mostafa Ahmed, Zeyad Emad, Eslam Amer et al. [1]	The supervised machine learning ways to detect and prevent cyberbullying.	SVM, NN
2.	Kelly Reynolds, April Kontostathis and Lynne Edwards et al. [2]	Detect language patterns used by the bullies.	C4.5 decision tree algorithm, instance-based learning, J48 algorithm
3.	Cynthia Van Heel, Gilles Jacobs, Chris Emmery, Bart Desmet, Els Lefever, Ben Verhoeven, Guy De Pauw, Walter Daelemans, VeAronique Hoste et al. [3]	Use of the linear support vector machines for automatic detection of cyberbullying in social media.	SVM
4.	Ghada M. Abaido et al. [4]	Calls for smartphone applications, stricter legal actions and proactive measures	Cronbach's alpha formula
5.	Elizabeth Byrne and Lauren Pfeifer et al. [5]	School nursing practice used for the prevention of cyberbullying	-----
6.	Mohammed Ali Al-Garadi, Mohammed Rashid Hussain, Nawsher Khan, Ghulam Murtaza, Henry Friday Nweke, Ihsan Ali, Ghulam Mujtaba, Haruna Chiroma, Hasan Ali Khattak and Abdullah Gani et al. [6]	Feature selection algorithms and then using various machine-learning algorithms for prediction of cyberbullying behaviors.	AI Classifiers
7.	Maral Dadvar and Kai Eckert et al. [7]	Deep learning-based models for the detection of cyberbullying on social media and also by overcoming the conventional models.	CNN, LSTM, BLSTM, BLSTM with attention

8.	Rui Zhao and Kezhi Mao et al. [8]	Machine learning techniques that make automatic detection of bullying messages in social media possible	Semantic-Enhanced Marginalized Denoising Auto-Encoder (smSDA)
9.	Semiu Salawu, Yulan He, and Joanna Lumsden et al. [9]	Categorized the present approaches into 4 main classes, they are supervised learning, lexicon-based, rule-based and mixed-initiative approaches.	SVM, Naïve bayes
10.	Huma Jameel, Neelam Duhan and Sana Shakeel et al. [10]	Natural Language Processing (NLP) and Machine Learning techniques	-----
11.	Amanpreet Singh and Maninder Kaur et al. [11]	Content-based learning	Machine learning techniques
12.	L. P. D. Bosque and S. E. Garza et al. [12]	The Machine Learning techniques for consequently distinguishing cases of cyberbullying	-----
13.	Konstantia Barmapsalou, Tiago Cruz, Edmundo Monteiro and Paulo Simoes et al. [13]	Presents a joined criminal profiling and suspicious example recognition technique	Neural and Neurofuzzy techniques
14.	Mohd Fazil and Muhammad Abulaish et al. [14]	Hybrid approach, an amalgamation of feature categories such as metadata-based, content-based and interaction-based features.	Random forest, decision tree and Bayesian network
15.	Tracy Vaillancourt, Robert Faris and Faye Mishna et al. [15]	Reviewed the ongoing writing on cyberbullying and its impacts on deceived youth, distinguishing key focuses	-----

3. CONCLUSION

Various Machine-learning techniques and its application were studied and reviewed. The Machine Learning algorithms were applied in the data sets. Thus, various machine-learning techniques provide an efficient way to detect cyberbullying.

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